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Online entertainment and exports: Evidence from a Chinese video game☆

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ABSTRACT

Online entertainment, represented by online videos, social media, and video games, have become an integral part of people's daily lives. How does online entertainment influence international economic activities and international relations? This paper investigates the impact of the late-2020 global success of a Chinese video game—Genshin Impact—on China's product exports, aiming to provide empirical insights into this question. Using monthly trade data and a Difference-in-Differences approach, we find that Genshin Impact had a significant positive effect on China's product exports. Regarding potential mechanisms, we present evidence suggesting that Genshin Impact led to a more favorable perception of China among foreign residents, thereby enhancing China's national image. This study provides novel evidence on how video games can influence international trade and national image, thereby advancing research in online entertainment, international trade, and international relations.

1. Introduction

The rapid development of online entertainment, represented by online videos, social media, and video games, has made them an indispensable part of people's lives. A prominent characteristic of online entertainment is its near-independence from temporal and spatial constraints. With just a click of a mouse or a tap on a screen, individuals can watch videos, interact with online communities, or launch a game—regardless of the geographical distance involved. Naturally, online entertainment has generated substantial profits for the companies behind them. However, many important questions remain unanswered. How does online entertainment impact the economies of the countries where relevant companies are based? What are its effects on international economic activities? How do it influence international relations? Addressing these questions is crucial for understanding the broader economic and social implications of online entertainment. Unfortunately, our knowledge in this area remains limited, and existing research lacks rigorous empirical evidence.

In this paper, we examine the impact of the late-2020 global success of a Chinese video game, Genshin Impact, on China's product exports, aiming to provide empirical insights into the aforementioned questions. Video games are among the most popular forms of

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entertainment. According to *Newzoo's Global Games Market Report 2024*, the number of global video game players is projected to grow by 4.5% in 2024, reaching 3.42 billion (Buijsman et al., 2024). Moreover, *Bain's 2024 Gaming Report* highlights that "..... the video game market continues to grow in all age groups, taking share of entertainment time and revenue from other media. The global market for video game revenue reached $196 billion in 2023, more than the combined revenue from streaming video ($114 billion), streaming music ($38 billion), and global box-office receipts ($34 billion)." Additionally, survey results from *Bain's 2024 Gaming Report* indicate that 52% of respondents play video games regularly, and nearly 80% of teenagers (aged 2 to 18) are gamers, dedicating 30% of their entertainment time to gaming (Christofferson et al., 2024).

Genshin Impact is a video game developed by miHoYo, a game company headquartered in Shanghai, China (for more details, see Section 2.1). Since its official release in late September 2020, *Genshin Impact* quickly gained global popularity and achieved remarkable success. To the best of our knowledge, Genshin Impact is the first truly globally recognized Chinese video game—one developed by a China-based company with a production team led by Chinese developers. In terms of revenue, Genshin Impact set multiple records for the fastest earnings milestones in mobile gaming history: surpassing $100 million in less than 14 days (Obedkov, 2020), $1 billion in under six months (Chapple, 2021), and $5 billion within 40 months (Obedkov, 2024). According to Statista, Genshin Impact has accumulated more than 202 million lifetime downloads as of November 2024. It is important to note that both revenue and download figures here refer only to the mobile platform, as data from other platforms remain undisclosed. We obtained data on Genshin Impact's monthly active players from ActivePlayer and plotted the trend in Appendix Fig. A1. As shown, since its official launch, the game's average monthly player count and all-time peak have continued to rise, eventually stabilizing at approximately 60 million and 17 million, respectively.

The popularity of Genshin Impact is also evident on various video and social media platforms. For example, on YouTube, the official English account[1] for Genshin Impact has 7.84 million subscribers, with 96 videos surpassing 5 million views and 27 videos exceeding 10 million views. The total number of views for the top 10 videos amounts to 251.22 million, with a total of 4.086 million likes. According to data from Exolyt, the TikTok hashtag "#genshinimpact" has generated 20.4 million videos and accumulated 246.0 billion views. The hashtag "#genshin" has 8.9 million videos, with a total view count of 116.7 billion.[2][3] Furthermore, Genshin Impact has frequently appeared on many countries' Twitter trending lists. In both 2021 and 2022, Genshin Impact was the most discussed game globally on Twitter in terms of game-related conversations (Chadha, 2022; Krdzic, 2022).

The exceptional quality and outstanding performance of Genshin Impact have earned it numerous accolades. First, in terms of ratings, the aggregate review site Metacritic shows that Genshin Impact has scores of 82/100 for iOS, 84/100 for PC (Personal Computer), 81/100 for PlayStation4, and 86/100 for PlayStation5. OpenCritic, which aggregates scores from global gaming media, gives Genshin Impact a Top Critic Average of 81 and a Critics Recommend score of 87%. In addition to ratings, Genshin Impact has received various prestigious awards. For example, it won the Best Mobile Game award at The Game Awards 2021[4] and the Players' Voice award at The Game Awards 2022.[5]

The questions we address are: How did the global success of Genshin Impact influence China's participation in international trade? And how did it affect the relationship between China and other countries? Based on panel data at the China-destination country-month level and a Difference-in-Differences (DiD) approach, we find that Genshin Impact had a significant positive effect on China's product exports. According to our simple back-of-the-envelope calculations, during the study period of this paper (from April 2020 to June 2021), Genshin Impact contributed approximately 3% to the growth of China's exports. It is important to note that Genshin Impact is a video game, not a trade policy. Therefore, we cannot directly compare the trade effects of Genshin Impact to those of trade policies, as it would not be reasonable to expect a video game to generate trade effects similar to those of trade policies.

A natural question is: What are the potential mechanisms through which Genshin Impact promotes China's exports? Within the context of this paper, two unique aspects of China are particularly relevant. First, the developer of Genshin Impact, miHoYo, is a Chinese company. Second, during the study period, the game's highly popular in-game country, Liyue, is based on a real-world prototype of China (see Section 2.1).[6] miHoYo has incorporated numerous elements of Chinese culture into Genshin Impact (Cai et al., 2022; Ramadany et al., 2024; Wang, 2023; Zhu and Huang, 2024).[7] Jaya and Zeniarja (2024) conducted a text analysis of tweets related to Genshin Impact on Twitter. The results show that players generally hold positive attitudes toward the game. Similarly, Zhang et al. (2024) performed sentiment and semantic network analyses of video comments on Genshin Impact on YouTube. Their findings

[1] Genshin Impact also has official accounts in other languages on YouTube, such as Japanese, which were not included in the analysis above.

[2] There are also other topics related to Genshin Impact that were not included in the analysis.

[3] The data from YouTube and Exolyt are current as of March 18, 2025.

[4] https://www.thegameawards.com/rewind/year-2021

[5] https://www.thegameawards.com/rewind/year-2022

[6] For example, among the top 10 most-viewed videos on the official English Genshin Impact YouTube account, 4 are character promotional videos (PVs) featuring Liyue characters. Of the 27 videos with over 10 million views, 9 are centered around Liyue characters or themes. The data is current as of March 20, 2025. When compared to Mondstadt, which was released around the same time as Liyue, the higher view count for Liyue does not appear to be solely due to its earlier release.

[7] It is important to note that Genshin Impact incorporates many elements from the real world. Each in-game country is based on corresponding real-world cultures (see Section 2.1). Additionally, Genshin Impact uses universal themes such as adventure and bravery to address cultural differences between countries (Yang, 2024). The inclusion of Chinese cultural elements in Genshin Impact has had a positive impact. For instance, cultural symbols, storylines, and characters related to Chinese culture in Genshin Impact have enhanced player loyalty (Chen et al., 2023). The game's Chinese-style settings and core narrative, which reflect the spirit of Chinese storytelling, convey a positive image of China to global players (Fan, 2024).

suggest that the videos evoked positive emotional responses from viewers, such as joy and surprise. Viewers also experienced deeper emotions such as appreciation, identification, and anticipation.

We believe that Genshin Impact will enhance players' positive feelings toward China or reduce their negative perceptions of the country, which may spill over to the surrounding environment. Interactions between individuals or groups can influence individuals' thoughts, emotions, attitudes, or behaviors, a phenomenon known as social influence (Cialdini and Goldstein, 2004; Rashotte, 2007). A large body of research has found that people share information and opinions, thereby affecting the attitudes of others, both offline and online. A typical example is word-of-mouth communication regarding products and services (Berger, 2014; Chen and Xie, 2008; Cheung and Thadani, 2012; Dubois et al., 2016; Sundaram et al., 1998). We expect that Genshin Impact has led residents of other countries to develop more positive emotions toward China or reduce negative feelings about China. In other words, Genshin Impact has encouraged a positive change in how residents of other countries perceive China, thus improving the China's national image. Consumers' perceptions and attitudes toward the country-of-origin (COO)—including the image and reputation of the COO—significantly influence their purchasing decisions (Bilkey and Nes, 1982; Heslop and Papadopoulos, 2014; Papadopoulos and Heslop, 2002; Roth and Diamantopoulos, 2009). Empirical studies based on bilateral trade data have also confirmed that national image and reputation can promote international trade (Anwar et al., 2026; Chang et al., 2022; Didier, 2019; Dimitrova et al., 2017; Rose, 2016; Rose, 2019; Xia and Xu, 2023; Zhi et al., 2017).

Consistent with our hypothesis, in our empirical study, we provide suggestive evidence that Genshin Impact has improved China's national image. This improvement effect is primarily driven by a reduction in negative feelings from other countries, rather than an increase in positive emotions. For China, this represents pennies from heaven brought about by Genshin Impact, particularly given the significant discrimination and stigmatization China faced during the COVID-19 pandemic. There were numerous incidents of discrimination and stigmatization of China, Chinese people, and Asians during the pandemic. A typical example is former U.S. President Donald Trump's public use of the discriminatory and stigmatizing term “Chinese Virus” instead of the neutral “COVID-19.” This significantly fueled anti-Asian sentiment both online and in real-world events (Cao et al., 2023; Hswen et al., 2021).

To investigate the impact of Genshin Impact on China's exports, we designed an empirical strategy in three steps. In the first step, based on the UN Comtrade database, we constructed a panel dataset on China's exports at the China-destination country-month level. In the second step, we developed a DiD econometric model. In constructing the treatment variable, we relied on the special relationship between Genshin Impact and Sony. There is evidence indicating that Sony and miHoYo maintain a very close relationship, and that Sony has played an important role in the global promotion of Genshin Impact. Sony is a Japanese multinational conglomerate with a large gaming business, including one of the largest console gaming platforms, PlayStation, and several game studios. In Section 2.3, we discuss the special relationship between Genshin Impact and Sony, but here we provide a brief overview. First, several Sony employees have publicly expressed the close relationship among Sony, miHoYo, and Genshin Impact. Second, there exist distinctive commercial practices between miHoYo and Sony. For example, the substantial exclusive cooperation agreement signed between Sony and miHoYo—for a long period, in addition to Genshin Impact's mobile and PC platforms, the console platform of Genshin Impact is exclusive to the PlayStation. Third, we provide empirical evidence that Sony's Google search popularity is positively correlated with Genshin Impact's performance and Google search popularity. We used Sony's Google Trends Index (GTI) in various countries before the study period as the treatment variable for Genshin Impact. In the econometric model, we included a range of control variables and fixed effects, such as the severity of the COVID-19 pandemic in the destination country, the destination country fixed effects and the month fixed effects, as well as the interaction terms between pre-existing characteristics of the destination country and month fixed effects.[8] These control factors help us mitigate the potential interference from the COVID-19 pandemic, pre-existing characteristics of the destination country, and some unobservable factors.

In the third step, we discuss the validity of the empirical strategy used in this paper (see Section 3.3). First, we conducted an event study corresponding to the DiD econometric model. The results show no evidence of significant trend differences between the treatment and control groups. This provides supporting evidence that the DiD method satisfies the parallel trends assumption. Second, we excluded the potential interference from the pre-existing export trends of China and any possible unobserved omitted factors. (i) In our regression, we controlled for China's exports to destination countries before the study period. (ii) We conducted a placebo test by lagging China's monthly exports. (iii) We used the GTIs of other companies (Nintendo, Rockstar Games, Ubisoft, Supercell, and Toyota) as placebo tests for Sony's GTI. (iv) We used the exports of other countries (Japan, South Korea, Vietnam, and India) as a placebo for China's exports. (v) We performed a permutation test with random assignment on the treatment variable—Sony GTI (Sun and Wang, 2025). All of these results support the validity of the empirical strategy employed in this paper. Finally, we employ a direct exposure measure—the destination country–month level Genshin Impact GTI. Our empirical analysis shows that the popularity of Genshin Impact in the destination country significantly stimulates China's exports.

The empirical findings of this study are as follows. First, the baseline regression results show that after the global release of Genshin Impact, for every 1 unit increase in the destination country's Sony GTI, China's exports to that country increased by 0.186%. Based on a simple back-of-the-envelope calculations, during the study period, Genshin Impact contributed approximately 3% to the growth in China's exports.

Second, regarding potential mechanisms, we utilized the Pew Global Attitudes & Trends database on public sentiment toward China and the GDELT database. We found that Genshin Impact significantly reduced the number and proportion of respondents holding mild negative views of China in the Pew survey, though there was no significant change in those holding strong negative views.

[8] It is important to note that the months in this study include the year dimension, i.e. $t = \{202004, \ldots, 202012, 202101, \ldots, 202106\}$. For convenience, we will refer to these as “month” in the subsequent discussion.

Furthermore, Genshin Impact also significantly reduced the number of negative events involving each country directed toward China in the GDELT. This effect was observed both in terms of the severity of negative impacts and the tone of negative sentiments.

Third, we incorporated gravity factors into the analysis. We found no evidence that geographical distance significantly weakened the export promotion effect of Genshin Impact on China. On the contrary, we found evidence that cultural distance significantly attenuated the export-promoting effect of Genshin Impact on China. This aligns with the characteristics of Genshin Impact as a video game and the role of Chinese cultural elements within the game. Specifically, as a video game, Genshin Impact is not limited by geographic distance in its spread. However, cultural distance can inhibit the effectiveness of Chinese cultural elements.

Finally, we introduced a product dimension into the analysis. We found that Genshin Impact significantly promoted China's exports at the intensive margin, but the extensive margin effect was very small. Moreover, the export-promoting effect of Genshin Impact on China was primarily driven by the quantity effect. We found that the price effect was negligible.

This study makes several contributions to the existing literature. The first strand of related literature concerns the extensive body of work in international trade, with particular attention to studies on country image and trade. A representative research area is the COO literature. Existing studies show that consumers' perceptions and attitudes toward a product's COO—its image and reputation—substantially shape purchasing decisions (Bilkey and Nes, 1982; Heslop and Papadopoulos, 2014; Papadopoulos and Heslop, 2002; Roth and Diamantopoulos, 2009). For example, Hsieh et al. (2004) find that consumers' views of the producing country strongly influence their automobile brand choices. Regarding the trade effects of country image and reputation, the literature documents positive impacts on exports (Anwar et al., 2026; Chang et al., 2022; Didier, 2019; Dimitrova et al., 2017; Rose, 2016; Rose, 2019; Xia and Xu, 2023; Zhi et al., 2017). These studies suggest that improvements in an exporting country's image and reputation enhance consumer preferences in destination markets, stimulate demand, and mitigate prejudice. The context of our study—Genshin Impact's global popularity—aligns with the broader theme of cultural diplomacy. Rather than examining the direct effect of country image on exports, our paper investigates the trade effects of events that may shape country image. Representative China-related cases in this domain include "panda diplomacy" and the establishment of Confucius Institutes. Jin et al. (2022) and Longhi and Morelli (2025) provide empirical evidence that hosting giant pandas significantly increases trade between recipient countries and China. Lien et al. (2012) and Chen et al. (2025) show that Confucius Institutes contribute positively to China's exports. To the best of our knowledge, this is the first empirical study to examine both the trade effects and country-image effects of video games. We find that the global diffusion of video games can promote the home country's exports and enhance its national image. This study contributes to the international trade literature, particularly the strand on country image and trade, and provides new empirical evidence on how video games influence international trade and national image within the fields of international economics and international relations.

The second category of contributions relates to the rapidly growing literature on online entertainment, including social media, online videos, and video games and so on. Researchers have explored the effects of online entertainment on individuals using experiments, surveys, and statistical analyses, with a focus on aspects such as mental health, behavior, and addiction. For example, Braghieri et al. (2022) examined the staggered introduction of Facebook in U.S. universities and found that social media had a negative impact on students' mental health. A study closely related to ours is Bailey et al. (2021), which leveraged Facebook user and friend data to construct a measure of social connections between countries. Their findings suggest that stronger social ties between two countries lead to greater trade flows and that these social media-based connections significantly weaken the explanatory power of geographical distance and national borders on trade. Our study differs from Bailey et al. (2021) in that we focus on the trade effects of video games rather than social media. Additionally, we examine the impact of video games on national image. By investigating video games as a driver of international trade and international relations, this study provides new empirical evidence on the broader influence of online entertainment.

The third category of contributions relates to the literature on cross-cultural interactions. Some studies have examined the effects of cultural differences and cultural distance on trade (e.g., Felbermayr and Toubal, 2010; Gokmen, 2017; Liu et al., 2020). Leveraging the Chinese cultural elements embedded in Genshin Impact, our study provides empirical support for the idea that video games can serve as vehicles for cultural dissemination. It is important to clarify that we do not directly test whether video games can facilitate cross-cultural transmission. Moreover, the notion of video games as tools for cultural dissemination is not new in either academic research or the gaming industry (Cerezo-Pizarro et al., 2023). Instead, our study investigates the trade effects of video games within the broader context of cross-cultural interactions. We hope that our findings will inspire future research on related topics, such as the role of online entertainment in cultural dissemination and the trade effects of cultural dissemination.

The remainder of this paper is organized as follows. Section 2 provides background information, introducing Genshin Impact and its developer miHoYo, discussing the possible reasons behind Genshin Impact's success, and examining the relationship between Genshin Impact and Sony. Section 3 presents the data and empirical strategy, detailing the datasets and econometric models used in this study while also assessing the validity of our empirical approach. Section 4 reports the empirical results, including baseline regression outcomes, robustness checks, and mechanism analysis, as well as an investigation of the role of gravity factors and the introduction of product dimensions in the empirical analysis. Finally, Section 5 concludes the paper.

2. Background

In Section 2.1, we introduce Genshin Impact and its developer, miHoYo. In Section 2.2, we discuss the possible reasons behind Genshin Impact's success. In Section 2.3, we explore the relationship between Genshin Impact and Sony through various news reports and anecdotes. This background information will help in understanding our empirical strategy and the possible mechanisms through which Genshin Impact has contributed to China's export growth.

2.1. Genshin impact and miHoYo

Genshin Impact is a video game developed by miHoYo, a company founded in 2011 and headquartered in Shanghai, China.[9] Before the release of Genshin Impact, miHoYo was a small to mid-sized game developer in China with little global recognition. Appendix Fig. B1 presents the GTI for miHoYo, showing that before the launch of Genshin Impact, global searches for miHoYo were nearly nonexistent. The explosive success of Genshin Impact was unexpected. News indicate that the Genshin Impact's performance exceeded industry expectations as well as miHoYo's own projections. Haoyu Cao, one of miHoYo's founders and then-chairman, expressed surprise at the Genshin Impact's global success (GameLook, 2021). Appendix Fig. B2 illustrates the GTI of Genshin Impact. Prior to its release September 28, 2020, global search interest in the game was nearly nonexistent. Shortly after its official launch, Genshin Impact experienced a rapid surge in popularity, reaching its peak level of attention within a very short period. In the following years, the game sustained a relatively high level of global interest. This sudden rise in popularity helps eliminate a potential expectation bias in our empirical strategy: if miHoYo had been a globally recognized game developer before Genshin Impact's release, or if Genshin Impact had already been widely known before its official launch, players in various countries might have formed expectations about Genshin Impact, potentially leading to misleading empirical results.

In Fig. 1, we outline the development timeline of Genshin Impact. Two key points need clarification. First, the Open Beta of Genshin Impact was essentially its official release. Therefore, we designate October 2020 as the current treatment period, with the study period spanning from April 2020 to June 2021. Second, the First Test signified that the game's fundamental framework was established, making major changes unlikely.[10] As a result, we use the period from July 2019 to March 2020 to construct our treatment variable. However, our core findings remain robust across different timeframes, as detailed in Sections 3.2, 3.3, and 4.2.

Genshin Impact's key features include cross-platforms, Role-playing Game (RPG), Open World, GaaS (Game as a Service), F2P (Free-to-Play), and gacha. Cross-platforms means players can log into Genshin Impact on mobile (phones and tablets), PC, and console platforms,[11] with account data syncing across all platforms. RPG and Open World[12] refer to the game genre, meaning the core elements of Genshin Impact are the map, storylines, characters, and equipment. GaaS, F2P, and gacha are business strategies. This strategy differs from GaaP (Game as a Product, where players pay upfront for all content) in two key ways. First, Genshin Impact continuously updates its game content. The game is set in seven in-game countries: Mondstadt, Liyue, Inazuma, Sumeru, Fontaine, Natlan, and Snezhnaya.[13] Each in-game country corresponds to a major update, called a "big map," with subsequent updates every year. In addition to the major updates, Genshin Impact releases minor updates (small maps) every 42 days.[14] Second, the vast majority of Genshin Impact's content—maps and storylines—is free for all players (F2P). However, certain characters and equipment are not free and require players to spend money for a chance to acquire them through gacha. This is the Genshin Impact's primary revenue model. Importantly, these characters and items do not affect the player's experience of the Genshin Impact's maps and storylines. In other words, the characters and equipment that can be obtained through gacha are not necessary to enjoy the game's maps and storylines, and they are independent of the core gameplay. This business model is one of the key factors behind Genshin Impact's success, which we will discuss in more detail in Section 2.3.

Now, let's introduce the game world of Genshin Impact. The game draws heavily from real-world elements, with each in-game country having a corresponding real-world cultural prototype. Table 1 summarizes the seven in-game countries and their real-world counterparts, as well as the version release dates. It should be noted that the countries in Genshin Impact correspond to real-world cultures, not specific countries. Apart from Liyue and Inazuma, which directly correspond to China and Japan, the other in-game countries draw from multiple countries around the real world. During the period of this study, specifically Version 1, Genshin Impact introduced Mondstadt and Liyue. As discussed in the introduction, China has two distinctive characteristics in this context. First, miHoYo is a Chinese company. Second, the highly popular in-game country, Liyue, is based on China. miHoYo has incorporated a significant amount of Chinese cultural elements into Genshin Impact (Cai et al., 2022; Ramadany et al., 2024; Wang, 2023; Zhu and Huang, 2024).[15]

[9] https://www.mihoyo.com/

[10] It should be noted that the First Test, Second Test, and Third Test of Genshin Impact were Closed Beta Tests (CBT). Interested players had to register and be selected through a limited qualification draw. For more details, refer to the official announcements on the Genshin Impact website: https://ys.mihoyo.com/main/.

[11] Before November 20, 2024, the only console platform on which players could access Genshin Impact was PlayStation. On November 20, 2024, Genshin Impact was officially launched on Xbox.

[12] RPG refers to the aspect where players control game characters and weapons. Open World means that players have a high level of freedom in navigating the game map.

[13] https://www.youtube.com/watch?v=TAlKhARUcoY

[14] Within a major version, Genshin Impact also updates smaller maps related to other major versions, but these are fewer in number. The primary focus of updates in a major version is on the storyline and map content related to that specific major version.

[15] It is important to note that Genshin Impact incorporates many elements from the real world. Each in-game country is based on corresponding real-world cultures (see Section 2.1). Additionally, Genshin Impact uses universal themes such as adventure and bravery to address cultural differences between countries (Yang, 2024). The inclusion of Chinese cultural elements in Genshin Impact has had a positive impact. For instance, cultural symbols, storylines, and characters related to Chinese culture in Genshin Impact have enhanced player loyalty (Chen et al., 2023). The game's Chinese-style settings and core narrative, which reflect the spirit of Chinese storytelling, convey a positive image of China to global players (Fan, 2024).

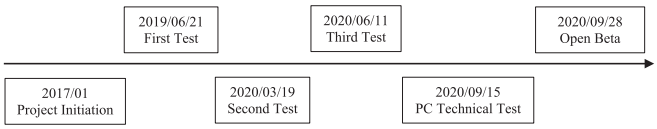

Fig. 1. The development timeline of Genshin Impact.

Table 1
Game countries in Genshin Impact.

Game countries	Archetypes of reality (culture)	Version	Start time
Mondstadt	Western Europe (Middle Ages)	Version 1	2020/9/28
Liyue	China	Version 1	
Inazuma	Japan	Version 2	2021/7/21
Sumeru	Persia, Arab, India, Egypt et al.	Version 3	2022/8/24
Fontaine	Western Europe (Morden Times)	Version 4	2023/8/16
Natlan	Africa, Latin America, Polynesia et al.	Version 5	2024/8/28
Snezhnaya	Eastern European		

Source: Genshin Impact website: https://ys.mihoyo.com/main/.

2.2. Possible reasons for the success of Genshin impact

In this section, we will discuss the possible reasons for the success of Genshin Impact, which will help in understanding the context of this study. The first reason for Genshin Impact's success is its inherent quality and the business strategy chosen by miHoYo. One of the main reasons Genshin Impact has been so beloved by players is its outstanding game mechanics, graphics, character designs, storyline, Chinese cultural elements, and sound effects (Ren, 2023). The ratings and awards mentioned in the introduction serve as direct proof of Genshin Impact's excellent quality.

Moreover, miHoYo has adopted a business strategy centered around GaaS, F2P, and gacha. As discussed in Section 2.2, Genshin Impact releases a minor update every 42 days. This development cycle ensures a continuous output of in-game content, which serves to both attract new players and retain existing ones. While this approach requires miHoYo to sustain substantial ongoing development costs—reportedly as high as $200 million annually (Landaverde, 2022)—the GaaS has successfully maintained Genshin Impact's exceptional popularity and generated substantial revenue. A comparative analysis of Genshin Impact and another major video game, Elden Ring,[16] using GTI illustrates this point. As shown in Appendix Fig. B3, global search interest in Elden Ring significantly exceeded that of Genshin Impact at the time of its release. However, within three months, Elden Ring experienced a sharp decline in search interest, falling below that of Genshin Impact and remaining lower thereafter. Even following the release of its DLC, Shadow of the Erdtree, on June 20, 2024, Elden Ring only temporarily surpassed Genshin Impact in search volume for approximately two months before declining once again. Crucially, the vast majority of Genshin Impact's game content is freely accessible to all players, consistent with the F2P. Revenue is generated primarily through the sale of specific characters and equipment, which are obtained via randomized draws—a system commonly referred to as gacha. Notably, players can experience the core gameplay elements of Genshin Impact, including its maps and storylines, without spending any money.

The combination of Genshin Impact's high production quality and miHoYo's chosen business strategies has significantly facilitated the game's rapid diffusion. Players face virtually no trial-and-error cost, as they can simply download the game and begin playing at no monetary expense. The high quality and F2P serve to attract potential users, the GaaS ensures player retention through ongoing content updates, and the gacha secures revenue. Together, these elements constitute a well-functioning commercial ecosystem. Importantly, these characteristics are consistent across all players. Therefore, we control for these factors using time fixed effects in our empirical analysis.

A second potential factor is the COVID-19 pandemic. In theory, the pandemic exerts two opposing effects on demand for Genshin Impact. On one hand, COVID-19 negatively affected both physical and mental health, while also causing significant economic disruption. These consequences may have reduced demand for Genshin Impact via health effects and income effects. On the other hand, during the pandemic, individuals voluntarily or involuntarily reduced outdoor activities and in-person social interactions. As a result, people had more discretionary time and a heightened desire for online entertainment and virtual socialization. This may have led to increased demand for Genshin Impact through a substitution effect. Notably, some media sources even described Genshin Impact as “the ultimate quarantine companion” (Tongco, 2022).[17] In addition, the COVID-19 pandemic had a substantial negative impact on

[16] Elden Ring was released on February 25, 2022, and quickly became one of the most prominent video games of its time, achieving both commercial success and critical acclaim. The game adopted a GaaP strategy, offering a one-time purchase with full access to the core content, and released only one major content update—a DLC—on June 20, 2024. Elden Ring received numerous industry awards, most notably the “Game of the Year” at The Game Awards 2022.

[17] This phenomenon is not unique to Genshin Impact. The COVID-19 pandemic played a significant role in the success of online games more broadly, largely because the majority of individuals were confined to their homes. Source: https://activeplayer.io/genshin-impact/.

international trade (Berthou and Stumpner, 2024; Bonadio et al., 2021). Therefore, the pandemic represents an important omitted variable in our empirical analysis. We address this concern by controlling for destination country-level COVID-19 conditions in our econometric model, as discussed in Section 3.2.

A third potential factor is the support provided by Sony. Some evidence suggests that Sony provided support for the global promotion of Genshin Impact. While we are unable to offer definitive proof—such as a formal contract or public announcement between miHoYo and Sony—we present suggestive evidence in Section 2.3. For firms engaged in international trade, customer acquisition is often constrained by numerous barriers (Lenoir et al., 2023). Firms typically employ marketing and advertising strategies to broaden their customer base, and may even adopt low markup in the early stages to gain market share (Fitzgerald et al., 2024). Prior to the launch of Genshin Impact, miHoYo lacked both experience and established channels for international promotion. Moreover, a global marketing strategy requires the firm to navigate a diverse set of market environments, as consumer preferences and tastes vary significantly across countries. In contrast, established game publishers possess long-standing industry expertise, reliable marketing networks, stable publicity channels, and accurate market intelligence—all of which were absent in the relatively young miHoYo. Sony can directly promote Genshin Impact on its own PlayStation console platform. In addition, as one of the largest and most reputable gaming companies globally, Sony can help miHoYo gain access to online and offline distributors in various countries, thereby facilitating stable business partnerships. Given the special relationship between Genshin Impact and Sony, we use the destination country GTI for "Sony" prior to the study period as a treatment variable. The rationale for this empirical strategy is discussed in Sections 3.3 and 4.2.

2.3. The relationship between Genshin impact and Sony

In this section, we discuss the special relationship among miHoYo, Genshin Impact, and Sony, which is crucial for understanding our empirical strategy. It is important to note that, to the best of our knowledge, neither miHoYo nor Sony has publicly disclosed any contractual agreements or issued formal announcements regarding their commercial collaboration. Consequently, we are unable to provide conclusive evidence regarding the specific ways in which Sony participates in the promotion of Genshin Impact, particularly because the details of such promotional activities constitute commercial secrets. In this section, we draw on collected materials and relevant news to present both textual and quantitative evidence. The textual evidence primarily consists of statements from miHoYo and Sony employees, as well as distinctive commercial behaviors observed between the two companies. The quantitative evidence relies on Sony GTI data, Genshin Impact's mobile performance data, and Genshin Impact GTI data, which together reveal a positive correlation between Sony and Genshin Impact. We acknowledge that none of these pieces of evidence are conclusive. However, taken together, they strongly suggest that the relationship between Sony and miHoYo is exceptionally close, and that Sony has played an important role in the global promotion of Genshin Impact.

Our search reveals that several Sony employees have publicly expressed the close relationship among Sony, miHoYo, and Genshin Impact. First, according to Reuters (Ye, 2022a), Tatsuo Eguchi, President of Sony Interactive Entertainment (SIE) Shanghai, stated that the success of Genshin Impact made Sony's management recognize the importance of Chinese games. Most importantly, he also noted that the partnership between Sony and miHoYo "is going strong" (Ye, 2022a). Second, according to an employee interview published on the official WeChat account of Sony Interactive Entertainment (Shanghai) Co., Ltd., Genshin Impact faced tremendous challenges before its release (including significant negative online pressure), yet Sony consistently and firmly supported miHoYo.[18] Third, Shuhei Yoshida—a key member of the PlayStation since 1993 and the President of SIE Worldwide Studios for Sony from 2008 to 2019—frequently mentioned Genshin Impact on Twitter. Notably, he once stated, "'Genshin Impact' from miHoYo has become one of my most anticipated games for 2020!" While this may be interpreted as a personal opinion, we argue that Yoshida's public endorsement, given his central role in Sony's gaming business, is indicative of broader institutional support from Sony toward Genshin Impact.

The distinctive commercial practices between miHoYo and Sony also reflect the closeness of their relationship. First, one piece of evidence for Genshin Impact's global promotion in support of Sony is its online marketing activity. For example, through manual searches, we identified that the official Twitter account @PlayStation, as well as several regional PlayStation accounts,[19] began posting tweets related to Genshin Impact as early as August 2019 and continued to do so on a regular basis. Some PlayStation accounts[20] posted less frequently or began promoting the game at a later stage (from September 2020 onward).

Second, surprisingly, for a long period of time, the only console platform on which players could access Genshin Impact was PlayStation. It was not until November 20, 2024, that the game was released on the Xbox, owned by Microsoft. According to Reuters (Ye, 2022b), miHoYo initially engaged in discussions with Microsoft regarding a potential collaboration for Genshin Impact, but no agreement was reached. Subsequently, in 2019, Sony and miHoYo entered into a partnership, and Genshin Impact was released globally one year later. Microsoft expressed regret over missing out on Genshin Impact (Ye, 2022b). As is well-known, competition among gaming consoles has been longstanding and highly intense, often referred to as the "Console War" (Harris, 2014). The rivalry

[18] https://mp.weixin.qq.com/s/u6JbLZgg-WuLMR13YhVp3g

[19] According to incomplete statistics, this includes regional PlayStation accounts such as PlayStation Europe, PlayStation Canada, PlayStation Latam (Latinoamérica), and PlayStation Brasil, among others.

[20] According to incomplete statistics, this includes regional PlayStation accounts such as PlayStation DE (Deutschland), PlayStation España, PlayStation UK, PlayStation Asia, PlayStation AU (Australia), PlayStation Ireland, PlayStation India, PlayStation France, PlayStation Portugal, among others.

between Sony and Microsoft in the console market is a classic example of this competition (Gamble, 2008). Media outlets have referred to the competition between PlayStation and Xbox as "The Battle for Control" (Richardson, 2023). Given the substantial exclusive cooperation agreement signed between Sony and miHoYo (Ye, 2022a), one of the returns for Sony's support of Genshin Impact's global promotion was miHoYo's agreement to make Genshin Impact a PlayStation exclusive for a period of time, meaning that the game would not be released on other console platforms.

Third, according to miHoYo's official website, miHoYo aims to build its own intellectual property (IP). As a result, miHoYo adopted a conservative approach to in-game crossovers, with the exception of Horizon Zero Dawn, a video game developed by Sony. miHoYo even created a playable character for Genshin Impact based on Horizon Zero Dawn's protagonist, Aloy, ensuring that her design aligned with Genshin Impact's artistic style, and offered her for free (Hogeweg, 2021). Additionally, Genshin Impact was specifically optimized for PlayStation platforms, granting PlayStation users unique advantages not available on other platforms.[21] For example, Genshin Impact on PlayStation 5 supports 4 K resolution with load times of only a few seconds. Furthermore, Genshin Impact developed an exclusive version of the Qingyun Peak map for PlayStation 5 players.[22]

Fourth, Genshin Impact won the Grand Award at the PlayStation Partner Awards for three consecutive years (2021 to 2023) and received the Excellence Award in 2024. In 2024, the PlayStation Partner Awards introduced a new category above the Grand Award—the Excellence Award—specifically designed for Genshin Impact. The criteria for this award were that the game must have achieved both commercial success and significant cultural impact within a three-year period, a definition that only one game, Genshin Impact, met. The purpose of this new award, in addition to recognizing the exceptional achievements of Genshin Impact, was also to free up the Grand Award category, which had effectively been monopolized by Genshin Impact (Wutz, 2024).

In addition to the textual evidence above, we also provide data-based quantitative evidence. If one of the reasons for the success of Genshin Impact is Sony's support, we should observe that the greater the Sony GTI of a country, the stronger the presence of Genshin Impact. To explore this, we present scatter plots of Sony GTI against Genshin Impact's mobile performance, as shown in Fig. 2. Panel 2 (a) displays the log of the average DAU for Genshin Impact in various countries during the study period, Panel 2(b) shows the log of the average MAU, Panel 2(c) represents the log of total downloads, and Panel 2(d) shows the log of total revenue. These data are obtained from Sensor Tower (see Section 3.1 for details). Figs. 2(a)-(d) demonstrate a positive correlation between Sony GTI and Genshin Impact performance: the larger the Sony GTI of a country, the higher the DAU, MAU, total downloads, and total revenue for Genshin Impact. To test the robustness of this relationship while accounting for population size, we present similar scatter plots in Appendix Fig. B4. These plots are identical to those in Figs. 2(a)-(d), except that the values on the vertical axes are divided by the total population of the country in 2019 and then logged. Appendix Fig. B4 shows that even after considering population size, the positive correlation between Sony GTI and Genshin Impact performance still holds.

As noted in Section 3.1, Sensor Tower data covers only 84 countries/regions and is limited to mobile platforms, excluding PC and console platforms. However, we believe these issues pose minimal threats to our identification strategy. First, Figs. 2(a)-(d) show that, at least on mobile platforms, we observe a positive correlation between Sony GTI and Genshin Impact performance. Second, we use Genshin Impact GTI to measure the popularity of Genshin Impact across different countries. In Figs. 2(e)-(f), we present scatter plots of Sony GTI against Genshin Impact GTI. Fig. 2(e) shows the topic "Genshin Impact" GTI for various countries, and Fig. 2(f) displays the term "Genshin Impact" GTI for those countries. Clearly, there is a positive correlation between Sony GTI and Genshin Impact GTI. To test the robustness of this relationship with the inclusion of other country characteristics, we present similar scatter plots in Appendix Fig. B5. The only difference is that we additionally control for the country-specific characteristics $\boldsymbol{X}_{c,2019}$ in Eq. (1). The results indicate that the positive correlation between Sony GTI and Genshin Impact performance, as measured by GTI, remains robust.

In summary, Fig. 2 shows that Sony GTI is positively correlated with Genshin Impact's mobile performance as well as with Genshin Impact GTI. Therefore, Sony GTI can, to some extent, capture the actual intensity of exposure to Genshin Impact on both mobile and PC platforms. For the PlayStation platform, due to data limitations, we are unable to provide empirical evidence. We acknowledge that this is a limitation of our empirical strategy and a potential avenue for future research. Nevertheless, the above analysis indicates that Sony and miHoYo maintain a very close relationship, and that Sony has played an important role in the global promotion of Genshin Impact.

Sony's support for Genshin Impact has resulted in substantial returns, particularly through the PlayStation exclusivity of the game. In terms of revenue, Genshin Impact won the Grand Award at the PlayStation Partner Awards for three consecutive years (2021 to 2023) and received the Excellence Award in 2024. This indicates that, over these four years, Genshin Impact was one of the highest-grossing games on PlayStation. Another potential benefit of the PlayStation exclusivity of Genshin Impact is its role in attracting and retaining more players on the PlayStation. Players drawn to Genshin Impact are likely to explore other PlayStation games, thus generating additional revenue and extending user engagement for Sony.

3. Data and empirical strategies

3.1. Data

The monthly export data at the HS6 (sixdigit Harmonized System) level for China's exports to destination countries from 2017

[21] For technical improvements and feature updates regarding Genshin Impact on PlayStation, please refer to the blog post published by Zhenzhong Yi, the Technical Director at HoYoverse Studio: https://blog.ja.playstation.com/author/zhenzhong/.

[22] https://blog.ja.playstation.com/2021/04/22/20210422-genshin/

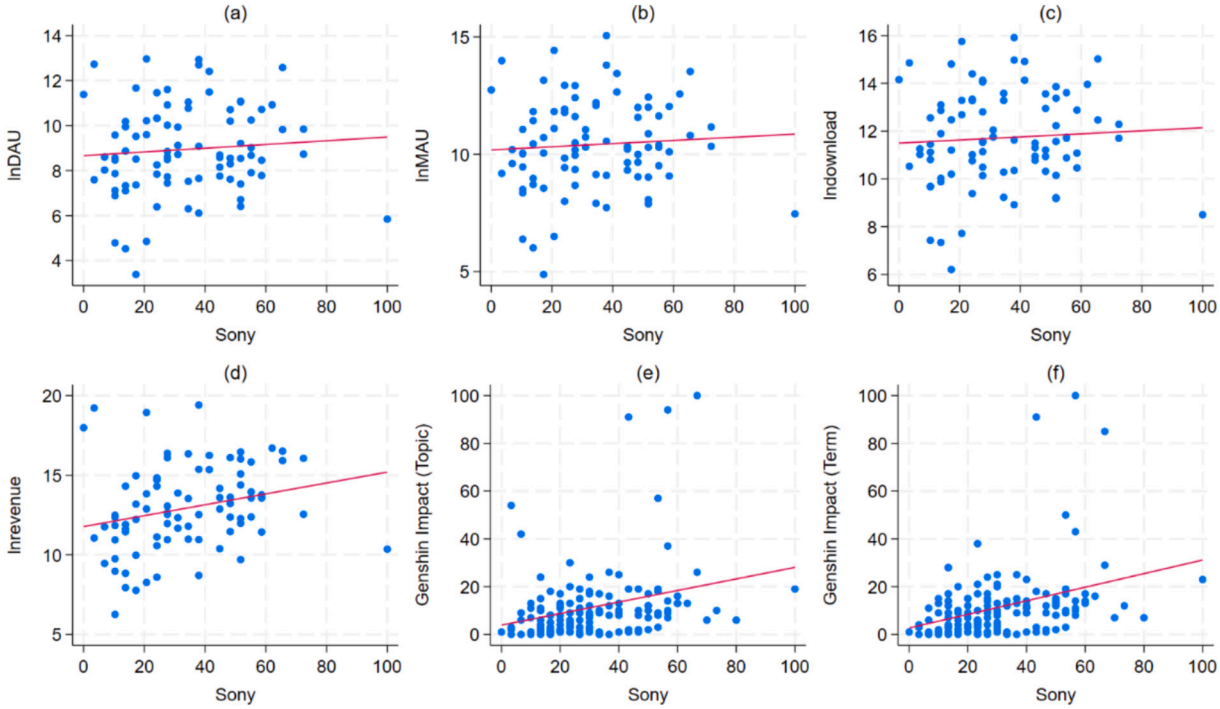

Fig. 2. The correlation between Sony GTI and Genshin Impact.
Note: The horizontal axis represents the topic "Sony" GTI of various countries prior to the study period (July 2019 to March 2020). The vertical axis of Fig. 2(a) shows the log of the average DAU of Genshin Impact in various countries during the study period (April 2020 to June 2021). The vertical axis of Fig. 2(b) shows the log of the average MAU, Fig. 2(c) shows the log of total downloads, and Fig. 2(d) shows the log of total revenue. The vertical axis of Fig. 2(e) represents the topic "Genshin Impact" GTI in various countries, and the vertical axis of Fig. 2(f) represents the term "Genshin Impact" GTI in those countries.

onwards are sourced from the UN Comtrade database. In the baseline empirical analysis, we aggregate the data from the China-destination country-HS6-month level to the China-destination-month level. In Section 4.5, we introduce the product dimension into the empirical analysis by conducting regressions at the China-destination country -product-month level. We also obtained monthly export data for Japan, South Korea, Vietnam, and India from UN Comtrade to serve as placebo exports for China during the same period.

To measure the presence of our objects of interest across countries, we utilized the Google Trends. Leveraging the vast coverage and data volume of Google search,[23] the Google Trends provides search intensity for a given object of interest across both time and geographical dimensions. To the best of our knowledge, Google Trends is the only data source available to us that allows for the measurement of the presence of our objects of interest at the country level. The Google Trends has been widely used in research (e.g., Borup and Schütte, 2022; Da et al., 2015; Preis et al., 2013; Vosen and Schmidt, 2011), demonstrating its excellent capabilities. The Google Trends offers two types of data: topic and term. Compared to terms, topics consolidate variations of the same object across different languages.[24] Therefore, unless otherwise specified, we use the topic type from the Google Trends database. For instance, in the case of Sony, we searched for the topic "Sony" in the Google Trends, setting the time period from July 2019 to March 2020, which provided us with the country-level GTI for Sony. This Sony GTI reflects the relative intensity of searches for Sony across different countries during the period from July 2019 to March 2020. In our empirical analysis, we normalized the raw GTI data to a scale of [0,100] to facilitate the assessment of effect magnitudes.[25]

The Covid-19 pandemic data is sourced from the Coronavirus Resource Center (CRC) database maintained by the Center for Systems Science and Engineering (CSSE) at Johns Hopkins University. The CRC collects and reports pandemic data from multiple sources, including confirmed cases and death tolls. Using the CRC database, we calculated country-month level pandemic indicators to

[23] According to Statcounter data, from 2017 to 2023, Google's global market share in search engines remained above 90%. Source: https://gs.statcounter.com/search-engine-market-share.

[24] For more information, refer to the Google Trends documentation: https://support.google.com/trends/answer/4359550.

[25] The normalized formula is $GTI_s = 100 \times (GTI - GTI_{min})/(GTI_{max} - GTI_{min})$. Here, the variable GTI refers to the country-level raw GTI, while GTI_{max} and GTI_{min} denote the maximum and minimum values of the raw GTI, respectively. According to the official Google Trends documentation, GTI is comparable across countries. The above transformation is monotonic and does not involve any change in units; therefore, the transformed GTI remains comparable across countries. In robustness checks, we show that using the raw GTI yields consistent results.

measure the extent of the Covid-19 impact. The annual country-level data is sourced from the World Bank's World Development Indicators (WDI) database.

We obtained the GDELT 1.0 Event Database from the GDELT Project. This database collects global broadcast, print, and web news on a daily basis, providing information on the time of the event, the two actors involved, and the action performed by Actor 1 upon Actor 2, among other details. Our focus is on events involving other countries (Actor 1) directed toward China (Actor 2). We fixed Actor 2 as China and calculated indicators at the Actor 1-China-month level. Specifically, we examine the Goldstein Scale, AvgTone, and the number of reports for each event.[26] The detailed definitions and construction methods for these indicators are provided in Section 4.3.

The performance data for Genshin Impact in various countries is sourced from Sensor Tower, covering 84 countries/regions. Performance metrics include DAU, MAU, downloads, and revenue. We use average DAU, average MAU, total downloads, and total revenue to measure the performance of Genshin Impact. It should be noted that, like other digital market insights platforms (e.g., Data.ai, formerly known as App Annie, which was acquired by Sensor Tower in March 2024[27]), Sensor Tower obtains app performance data from the iOS App Store and Google Play. Therefore, we only obtain the mobile performance data of Genshin Impact. To the best of our knowledge, there are currently no public sources available for obtaining performance data for the game on PC and console platforms across different countries. These issues will be discussed in Section 3.3.

Based on the UN Comtrade, Google Trends, CRC, and WDI databases, we constructed a panel dataset at the China-destination country-month level. We excluded certain special countries or regions, such as Vatican City, Hong Kong, Macau, and Taiwan. We also excluded small island countries. Additionally, we removed countries with missing Covid-19 data or those exhibiting abnormal mortality rates (i.e., countries with more than 100 confirmed cases but zero reported deaths).[28] Ultimately, we constructed a balanced panel consisting of 141 countries and 15 months (from April 2020 to June 2021), yielding a total sample size of 2115. During the study period, China's exports to these 141 countries accounted for over 80% of China's total exports. If Hong Kong, a key trade intermediary for China (Feenstra and Hanson, 2004), is excluded, China's exports to these 141 countries account for over 90% of its total exports.

3.2. Econometrics model

To study the impact of Genshin Impact on China's exports, we employ a DiD method. Considering the special relationship between Genshin Impact and Sony, we use the Sony GTI of each country prior to the study period as the treatment variable. In Sections 3.3 and 4.2, we will discuss and test the validity of using Sony GTI as the treatment variable in detail. The DiD model we construct is shown in eq. (1).

$$\ln export_{c,t}^{China} = \alpha + \beta_1 Sony_c \times Post_t + \beta_2 Sony_c + \beta_3 Post_t + \boldsymbol{Covid19}_{c,t} + \boldsymbol{X}_{c,2019}\theta_t + \delta_c + \theta_t + \varepsilon_{c,t} \tag{1}$$

In this model, c represents the destination country, and $t = \{202004, \ldots, 202012, 202101, \ldots, 202106\}$ represents the month. As introduced in Section 2.1, Genshin Impact began its First Test on June 21, 2019, and was officially released on September 28, 2020, with two playable maps: Mondstadt, inspired by Western Europe, and Liyue, inspired by China. On July 21, 2021, Genshin Impact introduced a new game map, Inazuma, inspired by Japan. Therefore, we designate October 2020 as the current treatment period, with the study period running from April 2020 to June 2021. This allows us to avoid any interference from the new version introduced with Inazuma.

The variable $\ln export_{c,t}^{China}$ represents the logarithm of China's export volume to country c in month t. Considering that Genshin Impact began its First Test on June 21, 2019, we use the topic "Sony" GTI, denoted as $Sony_c$, for the period from July 2019 to March 2020 in country c as the treatment variable. In Section 4.2, we show that changing the search period for $Sony_c$ or using the term "Sony" GTI yields robust results. $Post_t$ is a binary variable representing the treatment period, taking the value of 1 for months from October 2020 onwards (including October 2020), and 0 otherwise. $\boldsymbol{Covid19}_{c,t}$ is a set of Covid-19 indicators for country c in month t, including the number of confirmed cases, deaths, the case confirmation rate (i.e., the number of confirmed cases divided by the total population), the crude mortality rate (i.e., the number of deaths divided by the total population), and the case fatality rate (i.e., the number of deaths divided by the number of confirmed cases). Additionally, $\boldsymbol{X}_{c,2019}$ is a set of characteristics of country c in 2019, including the logarithm of GDP, the logarithm of GDP per capita, the logarithm of fixed broadband internet users, and the fixed broadband internet user density (per 100 people). We interact $\boldsymbol{X}_{c,2019}$ with time fixed effects θ_t to allow for different impacts of country c's characteristics in different months. Furthermore, we include country fixed effects δ_c and month fixed effects θ_t. The country fixed effects control for all time-invariant country characteristics, such as geographical distance and cultural differences between China and the destination

[26] Goldstein Scale: "(numeric) Each CAMEO event code is assigned a numeric score from -10 to +10, capturing the theoretical potential impact that type of event will have on the stability of a country." AvgTone: "(numeric) This is the average 'tone' of all documents containing one or more mentions of this event. The score ranges from −100 (extremely negative) to +100 (extremely positive)." For more details: http://data.gdeltproject.org/documentation/GDELT-Data_Format_Codebook.pdf

[27] https://sensortower.com/blog/data-ai-joins-sensor-tower

[28] The case fatality rate (CFR) of Covid-19 varies significantly across different countries and time periods. For example, in the early months of 2020, China's case fatality rate was 3.67%, while the United States had a case fatality rate of 3.43% (Karmakar et al., 2021; Verity et al., 2020). According to data from the World Health Organization, as of March 12, 2025, the global cumulative case fatality rate is 0.91%. Source: https://data.who.int/dashboards/covid19.

country. The month fixed effects control for all time-varying characteristics that are invariant across countries, such as global economic trends. $\varepsilon_{c,t}$ is the error term. We use robust standard errors clustered at the country level.

The coefficient of interest is β_1. β_1 represents the change in exports for the treatment group relative to the control group after the global release of Genshin Impact. The intensity of treatment is measured using the Sony GTI of each country prior to the study period. We expect the estimated coefficient $\widehat{\beta}_1 > 0$. For clarity, when the dependent variable is transformed using a logarithm or the inverse hyperbolic sine, we multiply the transformed dependent variable by 100 in order to directly interpret $\widehat{\beta}_1$ as a semi-elasticity. Detailed explanations can be found in the notes of each table and figure.

To preliminarily illustrate the relationship between Genshin Impact and China's exports, we present a scatter plot in Fig. 3(a) of the treatment variable $Sony_c$ and China's exports $\ln export_total_c$ during the study period (Berthou et al., 2024). Here, $export_total_c$ represents the total export value of China to destination country c during the study period, namely $export_total = \sum_t export_{c,t}^{China}$. It can be observed that there is a positive correlation between the Sony GTI and China's exports. Some may be concerned that Fig. 3(a) does not account for the population size of country c. To address this, we present a scatter plot in Fig. 3(b) of $Sony_c$ and the per capita China's exports ($\ln rexport_total_c$) to country c. Here $rexport_{total} = \sum_t export_{c,t}^{China} / population_{c,2019}$, $population_{c,2019}$ represents the total population of country c in 2019. Fig. 3(b) still shows a positive correlation between the Sony GTI and China's exports.

Appendix Table C1 provides descriptive statistics for the relevant variables. Appendix Fig. C1 to Appendix Fig. C6 display the performance and GTI of Genshin Impact in various countries during the study period. We find that Genshin Impact is distributed across major countries worldwide, and its presence seems to be positively correlated with the level of national development. In our regressions, we control for the pre-period characteristics of the destination countries to mitigate potential mechanical correlations. Moreover, Genshin Impact shows a higher presence in neighboring countries of China, such as Japan, South Korea, and Southeast Asia. This may be due to geographical proximity or cultural similarities. We will discuss this issue in Section 4.4.

3.3. Discussion

The validity of the DiD method relies on the parallel trends assumption under various control conditions and fixed effects. Specifically, it assumes that there is no significant trend difference between the treatment group and the control group prior to the shock. To provide evidence for the parallel trends assumption, we employ the common event study methodology used in the literature. Specifically, we construct the event study model as shown in Eq. (2).

$$\ln export_{c,t}^{China} = \alpha + \sum\nolimits_{t \neq 202009} \beta_t Sony_c \theta_t + \boldsymbol{Covid19_{c,t}} + \boldsymbol{X_{c,2019}} \theta_t + \delta_c + \theta_t + \varepsilon_{c,t} \tag{2}$$

We interact the treatment variable $Sony_c$ with a set of monthly dummy variables θ_t. The coefficient β_t represents the difference in outcomes between the treatment and control groups at time t. We use the month prior to the shock, September 2020, as the baseline. The other specifications in Eq. (2) are the same as those in Eq. (1). Fig. 4 displays the estimated coefficients $\widehat{\beta}_t$ and their 95% confidence intervals. The results in Fig. 4 suggest that, prior to the shock, specifically before October 2020, we find no evidence of a significant trend difference between the treatment and control groups. This provides evidence for the validity of our DiD methodology.

Some may be concerned that unobserved omitted factors or common trends could lead to spurious results, such as the finding that Genshin Impact boosts China's exports, when using Sony GTI as the treatment variable. Although we mitigate the issue of omitted variables as much as possible by including country fixed effects, month fixed effects, and a range of control factors in the regressions, we acknowledge that we cannot directly test the orthogonality of Sony GTI. We attempt to demonstrate that this issue is not severe under our identification strategy. On one hand, the event study results shown in Fig. 4 confirm that, under our empirical strategy, the treatment and control groups satisfy the parallel trends assumption. This rules out the concern that the results are driven by pre-existing trends in China's exports or differences in the export trends of China across different levels of Sony GTI prior to the shock (Noghanibehambari, 2025). On the other hand, we will conduct several robustness checks in Section 4.2, but here we provide a brief overview. First, we control for China's export to destination countries before the study period in the regression, and the results remain robust. Second, we conduct a placebo test by lagging China's exports. Third, we use the GTI of other companies (Nintendo, Rockstar Games, Ubisoft, and Toyota) as placebo for Sony GTI. (iv) We also use the exports of other countries (Japan, South Korea, Vietnam, and India) as a placebo for China's exports. (v) We perform a permutation test (Sun and Wang, 2025) where Sony GTI is randomly assigned. These tests effectively rule out the influence of omitted factors or common trends, such as regional economic development trends. Finally, We empirically examine the effect of the direct exposure measure—the destination country–month level Genshin Impact GTI—on China's exports, while ruling out potential confounding from the PlayStation. Detailed checks and discussions are provided in Section 4.2.

4. Empirical results

In Section 4.1, we present the baseline regression results. In Section 4.2, we conduct robustness checks. In Section 4.3, we provide some suggestive evidence of the mechanisms through which Genshin Impact promotes China's exports. It should be noted that these pieces of evidence are indicative. In Section 4.4, we explore the role of geographic gravity and cultural gravity between China and the destination countries. In Section 4.5, we introduce the product dimension and analyze the extensive margin and intensive margin of Genshin Impact's effect on China's exports, as well as the quantity effect and price effect.

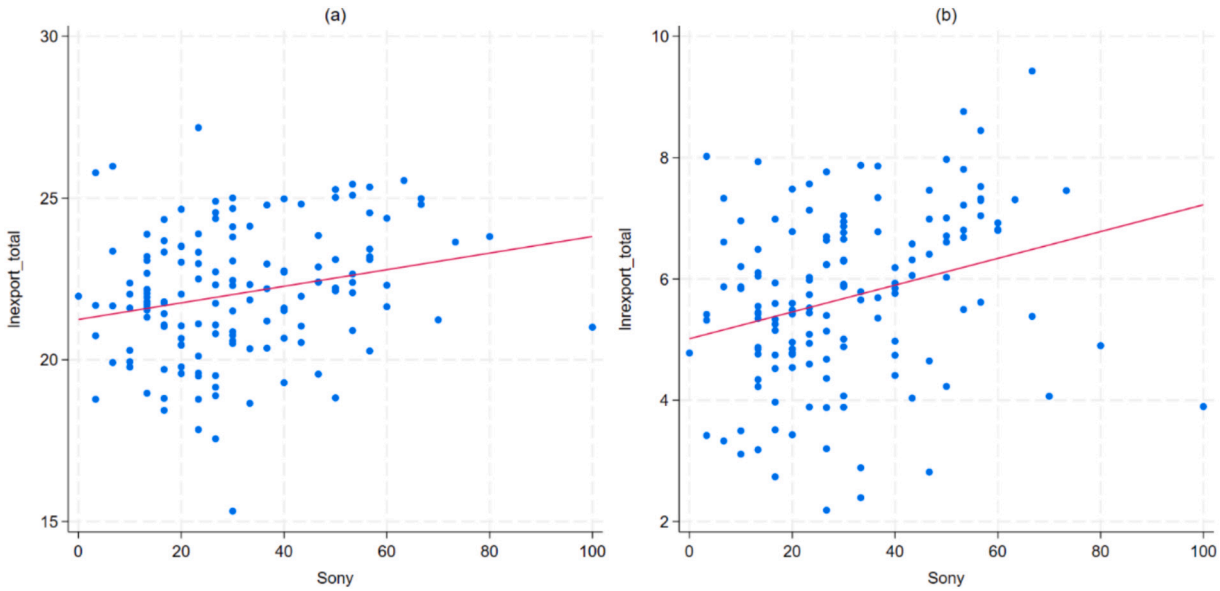

Fig. 3. The correlation between Sony GTI and China's exports.
Note: Fig. 3 is a scatter plot showing the correlation between Sony GTI and China's exports. The horizontal axis represents the topic "Sony" GTI for each country during the pre-study period (from July 2019 to March 2020). The vertical axis of Fig. 3(a) shows the log of China's total exports to each country during the study period (from April 2020 to June 2021). The vertical axis of Fig. 3(b) shows the log of per capita total exports from China to each country, with the population size being the total population of the destination country in 2019.

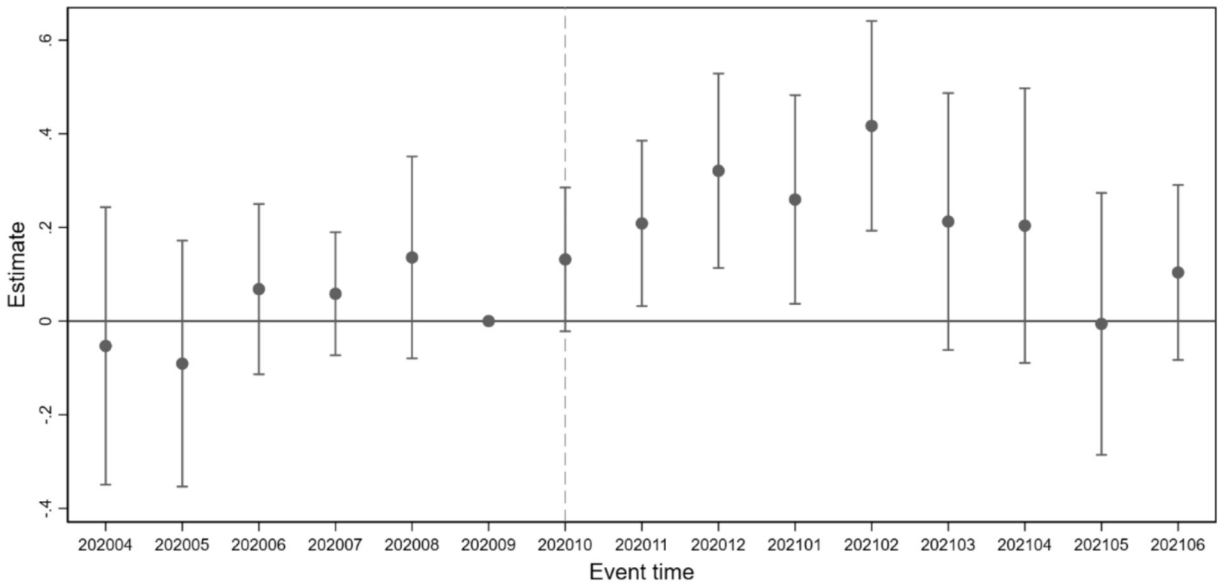

Fig. 4. Event study results.
Note: Fig. 4 displays the estimated coefficients $\widehat{\beta}_t$ from Eq. (2) along with their 95% confidence intervals. We multiply $\ln export_{c,t}^{China}$ in Eq. (2) by 100.

4.1. Benchmark regression

We present the benchmark regression results in Table 2. All columns in Table 2 control for country fixed effects and month fixed effects. Column (1) includes the COVID-19-related control variable $\boldsymbol{Covid19}_{c,t}$ from Eq. (1), but does not control for the interaction term of 2019 country characteristics and month fixed effects $\boldsymbol{X}_{c,2019}\theta_t$. As can be seen, the estimated coefficient is significantly positive at the 10% statistical level. Column (2) controls for the interaction term $\boldsymbol{X}_{c,2019}\theta_t$, but does not include the COVID-19-related control variable $\boldsymbol{Covid19}_{c,t}$ from Eq. (1). The estimated coefficient is significantly positive at the 5% statistical level. Column (3) includes all

fixed effects and control variables from Eq. (1) and represents our preferred specification. We find that the estimated coefficient is 0.186 and is significantly positive at the 5% statistical level. This means that after the global release of Genshin Impact, for each 1 unit increase in the destination country's Sony GTI, China's exports to that country will increase by 0.186%.

We now calculate the contribution of Genshin Impact to China's export growth on the back of an envelope. Suppose a country moves from the 25th percentile to the 75th percentile of the Sony GTI distribution. China's exports would increase by 0.186% $\times$ 26.7 $\approx$ 5%. From Fig. 2(e), we can observe that the unconditional correlation between Genshin Impact GTI and Sony GTI is 0.24. Therefore, the actual effect of Genshin Impact on China's exports is 1.2% (= 5% $\times$ 0.24). During the study period, China's exports grew by approximately 39.13%. This implies that Genshin Impact contributed about 3% (= 100% $\times$ 1.2% / 39.13%) to China's export growth. It should be noted that the result reflects a relatively local impact, rather than a general equilibrium effect. Furthermore, this calculation is on the back of an envelope. Genshin Impact is a video game, not a trade policy. We cannot compare the trade effects of a trade policy with those of Genshin Impact, as we would not expect a video game to generate trade effects similar to those of a trade policy.

4.2. Robustness checks

In the baseline regression, we use each country's topic "Sony" GTI during the pre-study period (July 2019 to March 2020) as the treatment variable. To assess the robustness of using GTI as a treatment variable, we vary the search time windows in the Google Trends database to construct alternative treatment variables. Specifically, we generate six alternative time periods: July 2019 to December 2019, January 2019 to December 2019, January 2018 to March 2020, January 2018 to December 2019, January 2017 to March 2020, and January 2017 to December 2019. Columns (2) through (7) of Appendix Table D1 report the corresponding results. Column (1) of Panel A in Appendix Table D1 replicates the baseline regression result, which corresponds to Column (3) of Table 2. In addition, Columns (1)–(7) of Panel B in Appendix Table D1 present results using the term "Sony" GTI instead of the topic "Sony" GTI as the treatment variable, based on the same set of alternative time windows. The results in Appendix Table D1 demonstrate that our findings are robust to changes in the search period and to using the term "Sony" GTI instead of the topic "Sony" GTI.

As discussed in Section 3.3, one potential concern is that unobserved omitted factors or common trends may bias our empirical results, leading to spurious findings when using Sony GTI as the treatment variable. To address this concern, we first conduct robustness checks by controlling for China's exports prior to the study period, followed by placebo tests. These exercises aim to demonstrate that such concerns are not severe in the context of our empirical strategy. We implement two approaches to control for China's pre-period exports. First, we compute the value of China's exports to each destination country in 2019 ($export_{c,2019}$), from 2018 to 2019 ($export_{c,2018-2019}$), and from 2017 to 2019 ($export_{c,2017-2019}$). After taking logarithms, we interact these values with month fixed effects to allow for time-varying impacts of China's pre-period exports. This method mirrors the use of $\boldsymbol{X_{c,2019}}\theta_t$ in Eq. (1). Regression results are reported in Columns (1) to (3) of Appendix Table D2. Second, for each month during the study period, we construct lagged export values from China. Specifically, we compute the export value from one year lagged, take the logarithm, and include it as a control variable, denoted as $\ln export_{c,t\ (L1)}$. We then compute the sum of export values from one and two years lagged, take the logarithm, and include it as an alternative control, also denoted as $\ln export_{c,t\ (L12)}$. Similarly, we calculate the sum of exports from one, two, and three years lagged, take the logarithm, and use it as a third version of $\ln export_{c,t\ (L123)}$. Regression results from this approach are reported in Columns (4) to (6) of Appendix Table D2. The results in Appendix Table D2 indicate that our findings remain robust after controlling for China's exports prior to the study period.

We now conduct a series of placebo tests to further alleviate concerns regarding omitted variables or common trends that might confound our estimates. First, we examine whether our main effect of interest holds for China's exports prior to the study period—that is, whether there exists a significant pre-trend. To do so, we use China's monthly export values from one year lagged as a placebo for the actual export values during the study period. Specifically, we replace the dependent variable in Eq. (1) with the previously constructed $\ln export_{c,t\ (L1)}$, which now reflects lagged export values. If a significant pre-trend exists, we would expect the placebo estimates to be significantly positive. Column (1) of Appendix Table D3 reports the regression results. We find no evidence that the placebo coefficient is significantly positive. Similarly, we replace the dependent variable in Eq. (1) with the previously constructed versions of $\ln export_{c,t\ (L12)}$ and $\ln export_{c,t\ (L123)}$ based on exports from earlier years. Results in Columns (2) and (3) of Appendix Table D3 show that the corresponding placebo estimates are statistically insignificant. Unlike the event-study-based parallel trend test presented in Fig. 4, the placebo tests in Appendix Table D3 allow us to examine the export trend over a longer historical window. The findings indicate the absence of significant pre-trends, lending further support to the validity of our identification strategy.

Second, a potential concern is that the Sony GTI might reflect some common trends, which could lead to spurious results. For instance, consumers might simultaneously increase their preferences for video games and Chinese products, or the Sony GTI might capture certain characteristics of the Japanese economy that are closely linked to China's economic performance. To address this, we construct a placebo for Sony. In response to the concerns mentioned above, Nintendo serves as an ideal placebo for Sony, given that Nintendo is also a globally recognized Japanese video game company.[29] We replace the treatment variable $Sony_c$ in Eq. (1) with the Nintendo GTI for the same time period (July 2019 to March 2020), using it as a placebo treatment variable. The regression results are reported in Column (1) of Appendix Table D4. We observe that the placebo estimate for the Nintendo GTI is not statistically significant.

[29] Nintendo's flagship games include the Super Mario Bros and The Legend of Zelda series and so on, and the company also owns the handheld game console platform, the Switch.

Table 2
Benchmark regression results.

	(1)	(2)	(3)
	ln*export*		
Sony×*Post*	0.157*	0.189**	0.186**
	(0.0816)	(0.0840)	(0.0832)
Covid19 controls	√		√
$\mathbf{X_{2019}}$ × Month FE		√	√
Country FE	√	√	√
Month FE	√	√	√
Observations	2115	2115	2115
R-squared	0.990	0.990	0.991

Note: The values in parentheses represent country level clustered robust standard errors. The variables *Sony* is absorbed by the country fixed effects. The variable *Post* is absorbed by the month fixed effects. In the regression, we multiply ln*export* by 100 to directly interpret the core estimated coefficient as a semi-elasticity. Covid19 controls include the number of confirmed cases, deaths, the case confirmation rate (i.e., the number of confirmed cases divided by the total population), the crude mortality rate (i.e., the number of deaths divided by the total population), and the case fatality rate (i.e., the number of deaths divided by the number of confirmed cases). $\mathbf{X_{2019}}$ includes the logarithm of GDP, the logarithm of GDP per capita, the logarithm of fixed broadband internet users, and the fixed broadband internet user density (per 100 people). *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Similarly, we use two other globally renowned video game companies as placebos for Sony—Rockstar Games from the United States and Ubisoft from France.[30] In addition, we use the globally renowned mobile game company Supercell from Finland as a placebo for Sony.[31] The regression results for these companies are reported in Columns (2) to (4) of Appendix Table D4. The placebo coefficients for Rockstar Games, Ubisoft and Supercell are also not statistically significant. Finally, we also consider Toyota, Japan's most important company and one of the largest automobile manufacturers in the world, as another placebo for Sony. Given Toyota's significant role in the Japanese economy, we examine its placebo effect. The results in Column (5) of Appendix Table D4 show that the placebo estimate for Toyota is not statistically significant either.

Third, there is a potential concern that regional trends could be driving the observed results. While we mitigate this issue through the use of various fixed effects and control variables, it cannot be fully ruled out. To address this, we construct a placebo for China. From the perspective of geographic proximity, Japan, South Korea, and Vietnam serve as ideal placebos for China, as they are all located in East Asia.[32] We use the export values of Japan, South Korea, and Vietnam to each country during the same period as placebos for China's exports. The regression results are reported in Columns (1) to (3) of Appendix Table D5. We find no evidence that the placebo coefficients for Japan, South Korea, and Vietnam's exports are significantly positive. Additionally, we use India—geographically adjacent to China and experiencing rapid industrial growth—as another placebo for China's exports. The results in Column (4) of Appendix Table D5 show that the placebo coefficient for India's exports is also not statistically significant.

Fourth, we conducted a permutation test with 500 random assignments of the Sony GTI treatment variable (Sun and Wang, 2025). Appendix Fig. D1 presents the density distribution of the placebo estimated coefficients from the 500 random assignments. The results in Appendix Fig. D1 show that the placebo estimated coefficients from the random assignments are centered around zero and are all positioned to the right of the baseline regression coefficient. This suggests that the findings of this study are unlikely to be driven by some coincidental factor.

We also conducted the following robustness checks. In Column (1) of Appendix Table D6, we use the raw Sony GTI as the treatment variable. The results indicate that standardizing the raw Sony GTI does not affect the robustness of our main effects. In addition, we include linear time trends at the destination country level. Column (2) of Appendix Table D6 shows that, after controlling for destination country-specific linear time trends, our main effects remain significantly positive.

Some may question the validity of using Sony GTI as a treatment variable for Genshin Impact. For instance, Genshin Impact is a multi-platform game, while Sony's PlayStation is a console platform. Moreover, Sony GTI not only reflects the gaming domain but is also influenced by brand recognition, consumer electronics consumption, and income levels. In addition, the release of the PlayStation 5 and PlayStation-exclusive titles during our study period could potentially affect the validity of Sony GTI as a treatment variable for Genshin Impact. We address these concerns from three perspectives. First, in Section 2.3, we provide evidence that Sony and miHoYo maintain a very close relationship, and that Sony played an important role in the global promotion of Genshin Impact. Second, in our empirical design, we include destination-country fixed effects, time fixed effects, and interactions between baseline economic

[30] Rockstar Games' flagship titles include the Grand Theft Auto series and the A Fistful of Dollars series, while Ubisoft is known for its Assassin's Creed series and Watch Dogs series.

[31] Supercell Games' flagship titles include Clash of Clans, Hay Day and Boom Beach.

[32] It is important to note that, from the perspective of cultural similarity and economic, Japan, South Korea, and Vietnam may not be ideal placebos for China. For instance, consumer preferences for China could spill over to Japan, South Korea, and Vietnam due to similar cultural characteristics. Additionally, the close economic links between China and these countries may result in spillover effects through economic channels. Therefore, the results here should be considered as heuristic references rather than definitive evidence. Fortunately, the placebo results suggest that the exports of Japan, South Korea, and Vietnam were not significantly positively affected by Genshin Impact.

characteristics of destination countries and time dummies. We also conduct multiple placebo tests and control for destination country specific linear time trends. These analyses help to mitigate concerns about omitted variables or common trends. Third, the release of the PlayStation 5 and many PlayStation-exclusive titles occurred after March 2020.[33] In our robustness checks, we repeatedly vary the time windows used to construct Sony GTI. The results show that our main effects remain robust across these periods. Since the releases of the PlayStation 5 and these games fall outside the Sony GTI search windows, we conclude that their launches pose minimal threat to the validity of Sony GTI as a treatment variable.

Finally, we conducted robustness checks, supplementing empirical results with measures based on direct exposure indicators.[34] We use the destination country–month level Genshin Impact GTI as the core independent variable to directly examine the effect of Genshin Impact popularity in destination countries on China's exports. To eliminate potential confounding from the PlayStation, we exploit Google Trends' functionality that allows simultaneous searches for multiple objects, where the resulting GTIs reflect the relative popularity between the search objects. Specifically, we simultaneously search for the topics "Genshin Impact" and "PlayStation" on Google Trends, yielding relative GTIs for Genshin Impact and PlayStation. These searches are conducted at the country level, producing destination country–month level GTIs.

We regress China's exports on the obtained Genshin Impact GTI, keeping all other conditions (control variables, fixed effects, and clustering level) the same as in the baseline regression (Column (3) of Table 2). The results are reported in Column (3) of Appendix Table D6. The regression shows that the Google search popularity of Genshin Impact has a significant positive effect on China's exports. In Column (4) of Appendix Table D6, we further control for PlayStation GTI. The estimated coefficients of Genshin Impact GTI in Columns (3) and (4) are nearly identical, which is expected, because we simultaneously searched for the topics "Genshin Impact" and "PlayStation". The resulting Genshin Impact GTI is measured relative to PlayStation's Google search popularity, meaning it has already been normalized by PlayStation GTI. Notably, the estimated coefficients of PlayStation GTI in Columns (3) and (4) are not significant, which alleviates concerns that PlayStation confounds our empirical design. Furthermore, because regressing China's exports directly on Genshin Impact GTI is highly endogenous, these results should be interpreted as merely suggestive. We prefer to rely on the DiD empirical design described in Eq. (1).

4.3. Potential mechanisms

In this section, we provide suggestive evidence on the potential mechanisms through which Genshin Impact may have promoted China's exports. As discussed in the introduction, there are two unique aspects of China that are relevant in this context. First, miHoYo is a Chinese company. Second, the highly popular in-game country, Liyue, is based on China. miHoYo has incorporated a significant amount of Chinese cultural elements into Genshin Impact (Cai et al., 2022; Ramadany et al., 2024; Wang, 2023; Zhu and Huang, 2024). These features may enhance players' favorable perceptions of China or reduce their aversion to the country, with potential spillover effects on the broader sociocultural environment. We hypothesize that Genshin Impact led to a positive shift in how people in other countries perceive China, thereby improving China's national image. To explore this possibility, we draw on data from the Pew Global Attitudes & Trends database and the GDELT database to provide suggestive evidence in support of this hypothesis.

We utilize the Summer 2020 and Spring 2021 waves of the Pew Global Attitudes & Trends database, both of which fall within our study period and bracket the global release of Genshin Impact. Given that data are available for only 13 countries, resulting in a total of 26 observations, we estimate a simple two-period DiD model. In Appendix D, we provide a detailed description of the Pew Global Attitudes & Trends database (Appendix Table D7 to Appendix Table D9, and Appendix Fig. D2), as well as the simple two-period DiD model. The regression results are reported in Appendix Table D10. Although we do not find evidence that Genshin Impact improved favorable perceptions of China, we do find evidence that it reduced unfavorable perceptions. This effect is primarily driven by a decline in mild aversion; there is no significant change in strong aversion. This result is intuitive—individuals who strongly dislike China are unlikely to revise their views solely because of a video game. Additional analyses based on the Pew Global Attitudes & Trends database are provided in Appendix D.

The survey data from the Pew Global Attitudes & Trends database are highly suitable for our research. However, a key limitation is the small number of countries surveyed with respect to perceptions of China. As such, the findings presented in Appendix Table D10 should be interpreted as suggestive evidence rather than conclusive proof. We now turn to the GDELT database to further investigate how Genshin Impact may have influenced the portrayal of China in media events. Specifically, we analyze two measures at the monthly country level: the total Goldstein Scale and the average Tone of events involving China. The Goldstein Scale captures the theoretical impact or intensity of an event, while the Tone index reflects the average sentiment associated with these events. Because both the Goldstein Scale and Tone can take negative values, we apply the inverse hyperbolic sine (IHS) transformation to each variable. We then replace the dependent variable in Eq. (1) with the transformed measures and estimate the specification accordingly. Columns (1) and (2) of Table 3 report the regression results. We find that Genshin Impact is associated with an increase in both the total Goldstein Scale and the average Tone of China-related events—suggesting that, relative to low-treatment countries, high-treatment countries exhibited stronger and more positive engagement with China in media narratives following the game's release. It is important to note, however, that these coefficient estimates are statistically insignificant and imprecise. Hence, while the direction of the estimates aligns with our

[33] For example, the release dates of the PlayStation 5, The Last of Us II, Ghost of Tsushima, and Spider-Man: Miles Morales were November 12, 2020, June 19, 2020, July 17, 2020, and November 12, 2020, respectively.

[34] We gratefully acknowledge the anonymous reviewers for their valuable suggestion to use direct exposure indicators and to account for potential confounding from the PlayStation platform.

theoretical expectations, the results should be interpreted with caution.

Subsequently, we examine the number of events involving each country directed toward China, along with disaggregated counts of events with positive or negative Goldstein Scale values, and events with positive or negative Tone values. Given that the dependent variables in this part of the analysis are event counts, we employ a Poisson Pseudo-Maximum Likelihood (PPML) estimator. Columns (3) to (7) of Table 3 present the regression results. We find evidence that Genshin Impact significantly reduced the number of China-related events across countries. This effect appears to be primarily driven by a decline in the number of events with negative Goldstein Scale values and negative Tone values. These findings are consistent with the results in Table 3 based on the Pew Global Attitudes & Trends database. While Genshin Impact did not significantly increase favorability toward China among foreign publics, it did reduce unfavorable perceptions. Analogously, while the game did not lead to a measurable increase in positively framed events involving China, it substantially decreased the number of negatively framed events. The reduction in negative events comprises two categories: (i) Events that reflect negative impacts on China. (ii) Events that express negative sentiment toward China. These results provide further suggestive evidence that Genshin Impact may have helped soften international perceptions of China by reducing negative narratives rather than amplifying positive ones.

Some literature has documented the positive impact of national image on exports. The most relevant study to the context of this paper is Anwar et al. (2026), who use the GDELT database from 2000 to 2014 to measure China's national image and empirically show that improvements in China's national image positively affect exports. In addition, using survey data from individual years and a limited number of countries (e.g., Pew and GlobeScan), Zhi et al. (2017), Chang et al. (2022), and Xia and Xu (2023) consistently find that enhancements in China's national image significantly promote China's exports. The positive effect of national image on exports is also confirmed in cross-country studies (e.g., Didier, 2019; Dimitrova et al., 2017; Rose, 2016; Rose, 2019). We have provided empirical evidence showing that Genshin Impact reduces negative sentiments toward China in other countries, thereby improving China's national image. Combining this finding—that Genshin Impact mitigates negative perceptions of China—with the literature confirming that national image promotes exports, we argue that the effect of Genshin Impact on enhancing China's national image constitutes an important potential mechanism through which it stimulates China's exports. This effect is particularly salient during our study period, when China faced discrimination and stigmatization due to the COVID-19 pandemic (Cao et al., 2023; Hswen et al., 2021). By reducing negative sentiment in other countries, Genshin Impact has made a non-negligible positive contribution to both the improvement of China's national image and the growth of its exports.

Overall, we provide suggestive evidence that Genshin Impact contributed to an improvement in China's international image. This improvement appears to be primarily driven by a reduction in negative sentiment from other countries, rather than an increase in positive sentiment. For China—particularly during the COVID-19 pandemic—this represents pennies from heaven brought about by the global success of Genshin Impact. It is important to note, however, that due to data limitations and imprecise regression estimates, these findings should be interpreted with caution. Nevertheless, we believe our results offer at least partial support for the hypothesis that Genshin Impact led to a more favorable perception of China among international audiences, thereby positively influencing the country's image. We acknowledge the need for more comprehensive data and improved methodologies to extend these findings. This remains a subject for further research.

Finally, the positive effect of Genshin Impact on China's exports may stem from an increased demand in destination countries for China's cultural and electronic goods. To examine this potential mechanism, we analyze the impact of Genshin Impact on China's

Table 3
Regression results based on the GDELT database.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Total	Average	Number				
	Goldstein Scale	Tone		Goldstein Scale		Tone	
			total	> 0	< 0	> 0	< 0
Sony×*Post*	0.691	0.0565	−0.00474*	−0.00393	−0.00578*	−0.00336	−0.00483*
	(0.957)	(0.267)	(0.00281)	(0.00259)	(0.00312)	(0.00264)	(0.00274)
Covid19 controls	√	√	√	√	√	√	√
$\mathbf{X}_{2019}$ × Month FE	√	√	√	√	√	√	√
Country FE	√	√	√	√	√	√	√
Month FE	√	√	√	√	√	√	√
Observations	1898	1898	1898	1898	1894	1898	1898
Pseudo/R-squared	0.419	0.347	0.9571	0.9423	0.9493	0.881	0.9569
Semi-elasticity			−0.473	−0.392	−0.576	−0.335	−0.482

Note: The values in parentheses represent country level clustered robust standard errors. The variables *Sony* is absorbed by the country fixed effects. The variable *Post* is absorbed by the month fixed effects. Columns (1) and (2) use the total Goldstein Scale and average Tone as the dependent variables, both of which are transformed using the inverse hyperbolic sine, and then multiply by 100. Columns (3)–(7) take as dependent variables the number of events involving each country directed toward China, along with disaggregated counts of events with positive or negative Goldstein Scale values, and events with positive or negative Tone values. The dependent variables are derived from the GDELT database. Columns (3) to (7) use the Poisson Pseudo-Maximum Likelihood (PPML) method. The last row of Table 3 reports the semi-elasticity estimates from the PPML estimation, i.e., $100\% \times (e^{\hat{\beta}} - 1)$. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

exports of cultural versus non-cultural goods, as well as electronic versus non-electronic goods. To identify cultural and electronic products, we use the UNESCO (United Nations Educational, Scientific and Cultural Organization) Framework for Cultural Statistics (FCS) and the UNCTAD (United Nations Conference on Trade and Development) ICT goods categories and composition. Both product classifications are at the HS6 level. Using HS6-level trade data from the UN Comtrade database, we aggregate China's exports to destination countries according to these classifications, obtaining export values for cultural goods, non-cultural goods, ICT goods, and non-ICT goods. We take the logarithm of these four types of export values and use them, respectively, as dependent variables in the baseline regressions. The results are reported in Table 4. They indicate that the positive effect of Genshin Impact on China's exports primarily arises from non-cultural goods and non-ICT goods.

We also conduct an analysis introducing the product dimension (for the detailed empirical design and econometric model, see Section 4.5). We define four product dummy variables: *Cultural goods*, *Non − cultural goods*, *ICT goods*, and *Non − ICT goods*. Building on the country–product–month-level regressions reported in Section 4.5, Table 5, we include these product dummy variables in the model. The results are reported in Appendix Table D11. We find that, although the extensive margin effects are primarily driven by non-cultural and non-ICT goods, in terms of the intensive margin, Genshin Impact significantly promotes exports across all four product categories.

Taken together, the results in Table 4 and Appendix Table D11 indicate that increased demand in destination countries for China's cultural and electronic products cannot fully account for the positive effect of Genshin Impact on China's exports. This finding lends partial support to the core mechanism proposed in this paper—that Genshin Impact enhances China's national image. Of course, these results do not imply that there is no direct spillover effect of Genshin Impact on demand for cultural and electronic products. The plausible interpretation is that the effect of Genshin Impact on improving China's national image plays a non-negligible role.

4.4. The role of gravity

In this section, we will analyze the role of gravity factors between China and the destination countries. We focus on both geographical and cultural gravity, aiming to provide further evidence for our core findings and potential mechanisms. One key feature of Genshin Impact is that it is F2P. With the exception of certain characters and weapons, players can experience almost all content in Genshin Impact without any cost. Genshin Impact is a video game. Unlike physical goods, miHoYo does not incur costs related to transportation, warehousing, or retail for Genshin Impact. Unlike traditional services or software, such as copyright-protected movies or commercial software, miHoYo does not impose restrictions on the distribution of Genshin Impact. Players only need to pay minimal costs for internet and electricity to easily download and experience the game. Moreover, platforms, such as Google Search, YouTube, and Twitter, which are used for search, video streaming, and social interaction, are not constrained by geographical distance. Therefore, we expect that geographical distance may not significantly weaken the role of Genshin Impact in promoting China's exports.

We obtained four bilateral distance indicators and one bilateral adjacency indicator from the CEPII database (Conte et al., 2022) to measure the geographical gravity between China and destination countries. The specific definitions of these indicators are provided in the note of Appendix Table D12. We interacted the five geographical gravity indicators with the DiD term $Sony_c \times Post_t$ in eq. (1), respectively. The regression results are reported in Appendix Table D12. The results in Appendix Table D12 show that, for all the geographical gravity indicators, we found no evidence suggesting that the promotional effect of Genshin Impact on China's exports varies significantly with changes in bilateral geographical gravity. This confirms our hypothesis that geographical distance does not significantly weaken the role of Genshin Impact in promoting China's exports.

In contrast to geographical distance, we believe that cultural distance may weaken the promotional effect of Genshin Impact on China's exports. Hofstede (2001) argues that culture is a broad tendency to prefer certain states of affairs over others. Residents of countries that are culturally more distant from China may find it more difficult to be moved by Genshin Impact and may struggle to understand the underlying messages expressed by the game's creators. To measure bilateral cultural distance, we use the 6-D model of national culture index provided by Hofstede (2011).[35] We followed the method outlined by Gorodnichenko et al. (2024) to measure cultural distance by calculating the absolute value of the difference in indices between China and other countries. Specifically, we first compute the absolute value of the differences for the six dimensions and then take the average. Given that some countries lack certain dimensions data, we generated two cultural distance indicators. The first does not require all six dimensions to be available, while the second requires all six dimensions to be present. We interacted the cultural distance indicators with the DiD term $Sony_c \times Post_t$ in eq. (1), respectively. The regression results are reported in columns (1) and (2) of Appendix Table D13. The results in Appendix Table D13 indicate that we found evidence suggesting that as bilateral cultural distance increases, the promotional effect of Genshin Impact on China's exports significantly diminishes. Additionally, we referred to the method of Kogut and Singh (1988) to calculate cultural distance. The specific construction method is outlined in the note of Appendix Table D13. The regression results are reported in columns (3) and (4) of Appendix Table D13. We found that cultural distance weakens the promotional effect of Genshin Impact on China's exports, although the estimates are not precise.

A natural question arises: which of the six dimensions contribute to weakening the promotional effect of Genshin Impact on China's exports? We interacted the absolute value of the differences in the six dimension indices with the DiD term $Sony_c \times Post_t$ in eq. (1), respectively. The regression results are reported in Appendix Table D14. The results in Appendix Table D14 indicate that the weakening effect of cultural distance is primarily driven by cultural differences in the IVR (Indulgence) dimension. Further inquiry into this

[35] The six dimensions are PDI (Power Distance), IDV (Individualism), MAS (Masculinity), UAI (Uncertainty Avoidance), LTOWVS (Long-term Orientation) and IVR (Indulgence). For more details: https://geerthofstede.com/

Table 4
Regression results based on cultural and ICT goods.

	(1)	(2)	(3)	(4)
	ln*export*			
	Cultural goods	Non-cultural goods	ICT goods	Non-ICT goods
Sony×*Post*	0.0924	0.198**	0.247	0.185**
	(0.118)	(0.0899)	(0.157)	(0.0820)
Covid19 controls	√	√	√	√
$\mathbf{X_{2019}}$ × Month FE	√	√	√	√
Country FE	√	√	√	√
Month FE	√	√	√	√
Observations	2115	2115	2115	2115
R-squared	0.990	0.990	0.991	0.991

Note: The values in parentheses represent country level clustered robust standard errors. The variables *Sony* is absorbed by the country fixed effects. The variable *Post* is absorbed by the month fixed effects. In the regression, we multiply ln*export* by 100 to directly interpret the core estimated coefficient as a semi-elasticity. Cultural goods and Non-cultural goods are based on the HS6-level list of cultural products provided by the UNESCO (United Nations Educational, Scientific and Cultural Organization) Framework for Cultural Statistics (FCS). ICT goods and Non-ICT goods are based on the HS6-level ICT product list provided by UNCTAD (United Nations Conference on Trade and Development) ICT goods categories and composition. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 5
The analysis results of introducing the product dimension.

	(1)	(2)	(3)	(4)	(5)
	export	*export01*	ln(*export* > *0*)	*quantity*	ln*price*
Sony×*Post*	0.00245	0.000607	0.447*	0.00875***	0.0714
	(0.00171)	(0.000545)	(0.257)	(0.00181)	(0.0507)
Covid19 controls	√	√	√	√	√
$\mathbf{X_{2019}}$ × Month FE	√	√	√	√	√
Country-product FE	√	√	√	√	√
Product-month FE	√	√	√	√	√
Observations	5,293,904	5,309,880	3,074,913	5,248,795	3,047,716
Pseudo/R-squared	0.97	0.577	0.796	0.9755	0.893
Semi-elasticity	0.2453			0.8788	

Note: The regression data in Table 5 is at the level of China-destination country-product-month. The values in parentheses represent country level clustered robust standard errors. The variables *Sony* is absorbed by the country-product fixed effects. The variable *Post* is absorbed by the product-month fixed effects. Columns (1) and (4) use the Poisson Pseudo-Maximum Likelihood (PPML) method. The last row of Table 5 reports the semi-elasticity estimates from the PPML estimation, i.e., $100\% \times (e^{\hat{\beta}} - 1)$. In columns (3) and (5), we multiply ln*(export > 0)* and ln*price* by 100 to directly interpret the core estimated coefficient as a semi-elasticity. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

matter goes beyond the scope of this study and is left for future research.

4.5. The analysis of introducing the product dimension

As the final part of our empirical analysis, we introduce the product dimension. Introducing the product dimension allows us to investigate both the extensive margin and intensive margin of Genshin Impact in promoting China's exports. Specifically, we aim to explore whether Genshin Impact increases the probability of China's exports or enhances the intensity of exports, or both. It also allows us to examine the quantity and price effects of Genshin Impact on China's exports, i.e., whether Genshin Impact increases the quantity of China's export products or raises the prices of export products, or both. To achieve this, we construct a panel dataset at the level of China-destination country c-product p-month m, and build the econometric model as shown in eq. (3).

$$\mathrm{Y}_{\mathrm{c,p,t}}^{\mathrm{China}} = \alpha + \beta_1 \mathrm{Sony_c} \times \mathrm{Post_t} + \beta_2 \mathrm{Sony_c} + \beta_3 \mathrm{Post_t} + \boldsymbol{Covid19}_{c,t} + \boldsymbol{X}_{c,2019}\theta_t + \delta_{cp} + \theta_{pt} + \varepsilon_{c,p,t} \tag{3}$$

Where $Y_{c,p,t}$ represents the indicators for the extensive margin, intensive margin, export quantity, and export price at the China-destination country c-product p-month m level. We control for destination country-product fixed effects δ_{cp} and product-time fixed effects θ_{pt}. Other specifications remain the same as in eq. (1). We define the product dimension p based on the finest product classification available in the UN Comtrade data, specifically HS6. For the extensive margin, we construct a binary variable $export01_{c,p,t}$, which is equal to 1 if there is export from China to destination country c for product p in month m, and 0 otherwise. For the intensive

margin, we calculate the non-zero export value for China-destination country c-product p-month m and take the natural logarithm, denoted as $\ln(export > 0)_{c,p,t}$. We also calculate the export quantity $quantity_{c,p,t}$ and average export price $price_{c,p,t}$ for China-destination country c-product p-month m. Note that since prices only exist when there is export, we directly take the logarithm of the price. Additionally, we calculate the total export value $export_{c,p,t}$ for China-destination country c-product p-month m to estimate the total effect. For the extensive margin, we estimate a linear probability model. For the intensive margin and price effects, we use OLS estimation. Given the large number of zero trade observations after introducing the product dimension, we apply the PPML method (Silva and Tenreyro, 2006) to estimate both the total effect and quantity effect. The regression results are reported in Table 5.

The results in column (1) of Table 5 show that, after the global release of Genshin Impact, for each additional unit Sony GTI increase in the destination country, China's exports to that destination country will increase by 0.2453%. This effect is slightly larger than the benchmark effect derived from Table 2. However, this estimate is imprecise (with a p-value of 0.152). The results from columns (2) and (3) of Table 5, which provide the decomposition of the total effect, indicate that Genshin Impact significantly promotes China's exports on the intensive margin. After the global release of Genshin Impact, for each additional unit Sony GTI increase in the destination country, China's exports will increase by 0.447%. However, we find that the effect on the extensive margin is very small. The decomposition results in columns (4) and (5) of Table 5 suggest that the promotional effect of Genshin Impact on China's exports is mainly driven by the quantity effect. After the global release of Genshin Impact, for each additional unit Sony GTI increase in the destination country, China's export quantity to that destination will increase by 0.8788%. However, we find that the price effect is very small.

After introducing the product dimension, we find that the promotional effect of Genshin Impact on China's exports is primarily driven by changes in the intensive margin. This finding is similar to the study by Fuchs et al. (2025). Fuchs et al. (2025) found that during the COVID-19 pandemic, China exported more masks to countries with strong bilateral political relations and experienced smaller price increases. This result was largely driven by changes in the trade intensive margin.

Finally, Appendix Figs. D3 to Appendix Figs. D7 report the event study results corresponding to each column of Table 5. In these figures, we replace the interaction term $Sony_c \times Post_t$ in Eq. (4) with interactions between $Sony_c$ and month fixed effects θ_t. In Appendix Fig. D3, we find that the event study coefficients for the total effect are statistically significant in the pre-treatment period (July and August 2020). In Appendix Figs. D5 and D6, we observe that the coefficients for the intensive margin and quantity effects are slightly significant in the pre-treatment period (August 2020). Therefore, we prefer to treat the analysis at the China–destination country–month level as our benchmark specification, while considering the analysis of introducing the product dimension as suggestive and exploratory.

5. Conclusion remarks

The internet and information technology have become deeply embedded in human society. Online entertainment—exemplified by online video, social media, and video games—now occupy a substantial share of individuals' leisure time. Existing understandings of their impact are primarily confined to the individual level. However, the broader implications of such entertainment on home country, international economic activities, and interstate relations remain largely unexplored. Empirical evidence on these fronts is particularly scarce. In this paper, we seek to fill this gap by employing a DiD approach and multiple data sources to examine how the global breakout success of a Chinese video game—Genshin Impact—in late 2020 affected China's exports. Our study aims to offer empirical insights into the aforementioned questions.

The empirical findings of this paper are as follows. First, Genshin Impact had a significant positive effect on China's exports. According to our simple back-of-the-envelope calculations, during the study period of this paper (from April 2020 to June 2021), Genshin Impact contributed approximately 3% to the growth of China's exports. Second, we find that Genshin Impact significantly reduced both the number and share of respondents in other countries who expressed mild aversion toward China. Additionally, it notably decreased the number of negative events involving each country directed toward China. Third, we find no evidence that geographic distance significantly weakens the export-promoting effect of Genshin Impact. In contrast, we do find evidence that cultural distance significantly attenuates this effect. This pattern is consistent with the nature of Genshin Impact as a video game and the role of Chinese cultural elements embedded within the game. Fourth, our results indicate that the export gains driven by Genshin Impact are mainly concentrated on the intensive margin and quantity effects, while the extensive margin and price effects are minimal.

This study contributes to the literature in international economics, online entertainment, and cross-cultural dissemination. Specifically, it provides novel evidence on how video games can influence international trade and national image, thereby advancing research in international trade, international economics, and international relations. Furthermore, the findings offer empirical support for the idea of video games as vehicles for cultural dissemination. We hope this study will inspire future research on related topics, such as the cultural dissemination effects of online entertainment and the trade effects of cultural dissemination.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.chieco.2026.102690.

Data availability

Data will be made available on request.

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